

Finite-Atlas Holonomy from Binary Self-Observation: A Reproducible S_3 Consensus Event in Boundary Atlases

SIQ / RelGauge Research Program

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Abstract

We present a finite computational framework in which gauge structure is not assigned as a primitive symmetry group but is read from the consistency of finite boundary observations. The primitive object is a finite transition system observed through a finite boundary atlas. A chart atlas induces local descriptions of unresolved fibers; transition maps between overlapping charts produce loop holonomies. Even very small finite systems can generate order-dependent reconciliation laws when multiple local chart disagreements overlap. We report a reproducible binary ($q = 2$) example in which information-preserving self-observation robustly generates finite C_2 chart holonomy and, at a critical birth window, two overlapping C_2 chart-cycle flips in the same namespace generate an exact nonabelian group S_3 . The nonabelian event is localized on a 17-microstate support inside a 512-state space. It is not a stable equilibrium sector; a local pulse-train audit classifies it as a one-tick critical flash. However, after freezing the transition table at the critical iteration, the event is algorithmically stable across the tested reconstruction family: 64/64 reconstructions recover exact target and support-level S_3 , with support Jaccard overlap 1.0, and a parameter-scale audit over 48 observation-method configurations also recovers exact target/support S_3 in every tested configuration. Negative controls are central to the result: lateral certified C_3/C_2 interfaces remain product-like, paired-return and moving-carrier audits do not produce nonvacuous S_3 , and endpoint/schedule-cut crossing-phase audits do not expose a deterministic three-phase carrier. The strongest supported conclusion is therefore not that Standard Model gauge has been derived, nor that S_3 is generic, but that finite, information-preserving self-observation can produce a localized, structural, nonabelian reconciliation event without inserting a nonabelian group by hand.

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1 Introduction

Gauge theory is usually formulated by postulating a symmetry group and requiring invariance under its local action. In the Standard Model, the gauge group is $SU(3) \times SU(2) \times U(1)$; in lattice gauge theory, group variables are placed on links; in fiber-bundle formulations, a structure group is specified and connections are then studied. This paper studies the converse direction in a finite setting: can nontrivial gauge structure be generated by finite self-observation, without specifying a nonabelian group in advance?

The framework considered here begins with a simple structural premise: every observation is made through a finite boundary. A finite boundary cannot resolve the entire microscopic state; it partitions the state space into visible classes and unresolved fibers. Multiple partial boundary windows yield overlapping local descriptions of the same fiber. Consistency of these local descriptions requires transition maps on overlaps. Around chart cycles these maps may fail to compose trivially, producing holonomy. This is the finite-atlas analogue of a gauge connection.

This situates the construction near, but not identical to, several established programs. Wilson’s lattice formulation and Kogut’s review show how gauge degrees of freedom can be represented finitely on links once a gauge group is chosen [6, 7]. Wheeler’s “it from bit” program motivates the information-theoretic direction [8]. Causal-set theory and Wolfram-style discrete rewriting models explore spacetime or dynamics from finite/discrete primitives [9, 10]. The distinction here is that the finite group is not specified on links in advance; it is extracted post hoc from chart-overlap reconciliation in a finite observer atlas.

The central result of this manuscript is a reproducible binary example. In a $q = 2$, $n = 9$ state space with 512 microstates, an information-preserving self-observation iteration produces a critical transition table. At iteration 8 of one reproducible random-full-permutation candidate, a local namespace (parent domain = 75, fiber = 7) contains two distinct order-two chart-cycle loop maps, (0 1) and (0 2), acting on a common three-label support. Since two transpositions sharing exactly one element generate S_3 , this is an exact nonabelian finite gauge closure. The microstate support of the event has size 17.

The paper makes three kinds of claims.

- (i) **Mathematical claims.** These are ordinary finite group statements, such as the fact that two overlapping transpositions generate S_3 .
- (ii) **Computational claims.** These are claims about the outputs of deterministic audit programs on specified inputs, seeds, and parameters. They are reproducible from the accompanying code and artifacts.
- (iii) **Interpretive claims.** These connect the computational findings to the idea of matter as multi-observer consensus. These are hypotheses and are deliberately stated more cautiously than the computational claims.

The strongest supported statement is:

A finite binary self-observation pipeline with information-preserving effective dynamics produces a localized 17-state structural S_3 consensus event, without inserting S_3 by hand.

The paper does not claim to derive the Standard Model, particle masses, or continuous spacetime. It does claim that a concrete nonabelian gauge-reconciliation event can arise from finite boundary consistency alone.

It is also useful to separate two uses of the word “gauge.” The computations demonstrate finite-atlas holonomy, including reproducible C_2 chart holonomy in information-preserving runs. Whether these finite holonomies become physical gauge fields in the Yang–Mills, lattice-gauge, or bundle-theoretic sense is a further scaling and interpretation problem, not a conclusion of this paper.

Boxed finite-boundary axiom

Finite-boundary axiom. Every observation is made through a finite boundary map $\pi : S \rightarrow \Sigma$, where $|\Sigma| < |S|$ for nontrivial observations. The inverse images $F_\sigma = \pi^{-1}(\sigma)$ are unresolved fibers. A finite observer may use multiple partial chart windows on these fibers, and descriptions on overlaps must be mutually reconcilable by finite transition maps. No external reservoir may absorb distinctions lost by the boundary; thus information-preserving effective dynamics is the axiom-compatible case, while representative collapse is an explicitly lossy control.

The computational pipeline below is one finite realization of this axiom. The manuscript’s main claim is constructive, not universal: it exhibits one reproducible finite system in which this pipeline yields a localized structural S_3 consensus event. It does not prove that every finite observer must generate S_3 , nor that the event is a particle or a Standard Model gauge sector.

2 Formal setup

2.1 Finite state systems

Let

$$S = \{0, 1, \dots, q - 1\}^n$$

be a finite state space. In the main result, $q = 2$ and $n = 9$, so $|S| = 512$. A microscopic dynamics is a map

$$T : S \rightarrow S.$$

The principal experiments use random full permutations as initial dynamics, so T is bijective. This choice is not a gauge input: it fixes a generic information-preserving initial transition table.

A finite observer does not access $s \in S$ directly. It observes a boundary projection

$$\pi : S \rightarrow \Sigma,$$

where Σ is a finite label set. For $\sigma \in \Sigma$, the fiber

$$F_\sigma = \pi^{-1}(\sigma)$$

contains microscopic states unresolved by the observer.

Definition 1 (Finite boundary atlas). *A finite boundary atlas A consists of a finite set of domains, each carrying a partition of a subset of S , together with charts on selected fibers and overlap transition maps between chart labels where charts describe common fiber states.*

The implemented atlas is generated by recursive proliferation. Starting from a simple boundary label, fibers are recursively refined according to predictive distinctions induced by the transition table. The procedure is finite because the atlas capacity, maximum domains, maximum charts per fiber, and maximum chart coordinate/support sizes are bounded.

2.2 Charts, overlap maps, and holonomy

Let F be a fiber, and let $c_i : F \rightarrow L_i$ be a chart assigning finite chart labels to states in F . On a region where two charts describe the same states, the induced transition map is a partial map

$$g_{ij} : L_i \dashrightarrow L_j.$$

A chart cycle is a sequence

$$c_{i_0} \rightarrow c_{i_1} \rightarrow \cdots \rightarrow c_{i_m} = c_{i_0}.$$

The composed loop map is

$$H = g_{i_{m-1}i_m} \circ \cdots \circ g_{i_0i_1}.$$

When H is a nontrivial permutation of chart labels, the atlas has finite holonomy. An order-two loop map is a C_2 holonomy.

Definition 2 (Same-namespace composition). *Two loop maps are composable for gauge closure only if they act on the same atlas namespace: the same candidate, iteration window, parent domain, and fiber label, or a certified equivalent support. This requirement avoids conflating labels from different observers or different fibers.*

2.3 Information-preserving self-observation

Given an atlas $A(T)$ built from a transition table T , the pipeline constructs an effective transition table

$$T_{\text{eff}} = \Phi(T)$$

by reading how T acts on atlas labels and lifting the finite effective dynamics back to sampled microstates. The next self-observation step uses

$$T \leftarrow T_{\text{eff}}.$$

Several lift names are implemented: `bijective`, `permutation`, `conservative`, `information_conserving`, and `representative`. The first four preserve the relevant information in the reference regime and produce identical gauge-generation statistics; the representative lift collapses fibers to representatives and destroys gauge generation in the reference test. Thus the empirical split is between information-preserving and information-destroying lifts, not between arbitrary tunable choices.

3 Finite group facts

The nonabelian result rests on an elementary theorem.

Lemma 1 (Overlapping transpositions generate S_3). *Let $X = \{a, b, c\}$ with distinct elements. Let $\tau_1 = (a\ b)$ and $\tau_2 = (a\ c)$. Then*

$$\langle \tau_1, \tau_2 \rangle \cong S_3.$$

Proof. Both generators are transpositions. Their product is a 3-cycle:

$$(a\ b)(a\ c) = (a\ c\ b)$$

up to composition convention. The subgroup generated by a transposition and a 3-cycle on the same three-element set contains all six permutations of X . Hence it is S_3 . $\square$

Proposition 1 (Finite-atlas S_3 certificate). *If two same-namespace chart-cycle loop maps are transpositions sharing exactly one chart label, then the finite chart-cycle closure contains exact S_3 .*

Proof. Restrict to the union of the three labels involved. Extend each transposition by the identity on labels outside its two-point support. The preceding lemma applies. $\square$

This proposition is the algebraic certificate used in the spatial chart-cycle closure and transposition-diversity audits.

4 Computational methods and reproducibility

4.1 State space and reference candidate

The reference nonabelian event occurs in a binary state space with $n = 9$ vertices. The target candidate is

```
random_full_permutation|inst=28|full_atlas|cap=32|seed=3221741.
```

The event occurs at atlas iteration 8, in parent domain 75, fiber label 7.

4.2 Deterministic seeding

A reproducible implementation must not use Python’s process-randomized built-in `hash()` in seed construction. The patched iteration code uses deterministic SHA1 hashing of profile strings. The relevant source file includes `hashlib` and the deterministic seed path. The repository tests include a deterministic-hash check and `grep` for process-randomized seed paths.

4.3 Frozen transition table

For multi-observer tests, the transition table at the critical iteration is saved as a `.npy` array. Downstream consensus audits load this frozen transition directly rather than replaying the entire iteration history. This removes ambiguity due to earlier non-deterministic hash behavior and makes observer reconstructions act on the same boundary configuration.

4.4 Audit modules

The paper relies on the following audit families.

- **Iterated fiber-atlas dynamics:** generates atlas sequences and tracks C_2 holonomy.
- **Spatial chart-cycle closure and transposition diversity:** composes same-namespace C_2 loop maps and classifies generated finite groups.
- **Critical birth-window and pulse-train audits:** locate local C_2 birth/rebound windows and classify flat, C_2 , and S_3 pulses.
- **Critical-event microscope:** recomputes the S_3 group, orbit size, and matched C_2 controls.
- **Multi-observer consensus:** rebuilds atlas observations from a frozen transition table and measures exact-namespace and support-level consensus.
- **Observation-method scale test:** varies observer profile, atlas capacity, and entropy threshold while measuring whether the S_3 support remains visible.

- **Negative-control audits:** certified inter-scale interfaces, paired returns, moving carriers, and crossing-phase tests.

4.5 Result provenance

Table 1 lists the principal result artifacts used in this paper. Filenames in the package and archive use the longer machine-readable names indicated in the repository manifest.

5 Results I: information preservation and C_2 generation

5.1 Lift-mode split

The reference lift-mode experiment used $q = 2$, $v = 9$, random full permutations, 29 instances, full-atlas profile, capacity 32, 12 atlas iterations, and seed control sufficient to reproduce the reference witness. The information-preserving modes `bijective`, `permutation`, `conservative`, and `information_conserving` produced the same summarized behavior: C_2 generation fraction $20/29 = 0.689655\dots$, maximum chart C_2 holonomy 116, and identical domain and chart-cycle counts in the reported run. The `representative` lift produced no C_2 gauge generation and collapsed to flat/fixed behavior in the reference comparison.

This result is important because it reduces an apparent implementation choice to an empirical equivalence class. In the tested regime, gauge generation is not tied to one named lift. It appears for the information-preserving class and vanishes for the representative collapse.

5.2 Broad C_2 generation

In the broad $v = 9$ run, C_2 generation occurs in 0.41944 of candidate histories, with final persistence 0.03148 and maximum chart C_2 holonomy 704. In the broad $v = 10$ run, C_2 generation rises to 0.58194, final persistence to 0.16528, and maximum chart C_2 holonomy to 800. Thus C_2 generation is robust and increases in the tested $v = 10$ run.

However, C_2 abundance is not the same as nonabelian closure. The $v = 10$ spatial closure audit finds 4812 same-namespace C_2 groups, all single or duplicate flips, with no shared-label pair, no component of size 3, and no S_3 . This is a key negative control: more C_2 does not automatically produce S_3 .

6 Results II: exact S_3 from spatial chart-cycle closure

6.1 The spatial closure event

The $v = 9$ spatial chart-cycle closure audit finds 3838 same-namespace C_2 -bearing groups. Of these, 3837 are single or duplicate C_2 flips, and exactly one has two overlapping transpositions. The exact S_3 fraction is therefore

$$\frac{1}{3838} \approx 2.6055 \times 10^{-4}.$$

The maximum generated group order is 6, and the maximum transposition component size is 3.

The witness occurs at

```
candidate = random_full_permutation|inst=28|full_atlas|cap=32|seed=3221741,
iteration = 8,
parent domain = 75,
fiber label = 7.
```

The nine nontrivial C_2 chart cycles in this namespace reduce to two distinct loop maps:

$$(0\ 2), \quad (0\ 1).$$

Thus, by Proposition 1, the local chart-cycle closure is exact S_3 .

6.2 Route redundancy

The event is not supported by a single accidental loop pair. The $(0\ 2)$ map appears along three chart routes,

$$1 \rightarrow 5 \rightarrow 8 \rightarrow 7, \quad 1 \rightarrow 5 \rightarrow 9 \rightarrow 7, \quad 1 \rightarrow 5 \rightarrow 10 \rightarrow 7,$$

while the $(0\ 1)$ map appears along six routes,

$$\begin{aligned} 5 \rightarrow 8 \rightarrow 7 \rightarrow 13, \quad 5 \rightarrow 9 \rightarrow 7 \rightarrow 13, \quad 5 \rightarrow 10 \rightarrow 7 \rightarrow 13, \\ 7 \rightarrow 8 \rightarrow 12 \rightarrow 13, \quad 7 \rightarrow 9 \rightarrow 12 \rightarrow 13, \quad 7 \rightarrow 10 \rightarrow 12 \rightarrow 13. \end{aligned}$$

This route redundancy strengthens the interpretation that the event is a structural chart-consensus feature of the frozen transition, not a single-cycle parsing accident.

7 Results III: critical birth-window provenance

The critical-birth-window audit restricts to namespaces in the first C_2 birth window, with generated-from-flat dynamics. It finds

$$1164 \quad C_2\text{-single/duplicate namespaces,}$$

and

$$1 \quad S_3 \text{ critical-birth namespace.}$$

The unique S_3 row has two distinct C_2 maps, a shared-label transposition pair, maximum transposition component size 3, generated group order 6, and a compression-rebound score approximately 359.50.

Matched controls in the critical-event microscope have group order 2 and orbit size 2, whereas the target has group order 6 and orbit size 3. The mean compression-rebound score of controls is approximately 25.05, far below the target's score.

The matched-control count in Table 3 is taken from the current verified microscope summary: $n_{\text{controls}} = 20$, all with group order 2, and mean control compression-rebound score 25.05. An earlier exploratory microscope run used 10 controls and is superseded by this verified artifact.

The pulse-train audit shows that this event is temporally local. In the local namespace, the sequence is

$$7 : \text{flat} \quad \rightarrow \quad 8 : S_3 \text{ flash} \quad \rightarrow \quad 9 : \text{flat}.$$

The S_3 duration is one atlas tick, with collapse mode `to.flat`. Thus the event is not a stable equilibrium sector but a local critical consensus flash.

8 Results IV: the 17-microstate support

The target support contains 17 microstate indices:

$$\{29, 30, 44, 47, 61, 62, 156, 159, 188, 191, 221, 222, 253, 254, 256, 284, 287\}.$$

Table 4 lists their binary representations and transition images under the frozen transition table at iteration 8.

Table 4: The 17 microstates of the structural S_3 support.

Index	Binary state	Image index	Hamming weight
29	000011101	33	4
30	000011110	34	4
44	000101100	48	3
47	000101111	51	5
61	000111101	65	5
62	000111110	66	5
156	010011100	160	4
159	010011111	163	6
188	010111100	96	5
191	010111111	99	7
221	011011101	225	6
222	011011110	226	6
253	011111101	129	7
254	011111110	130	7
256	100000000	329	1
284	100011100	288	4
287	100011111	291	6

The support has a visible $16 + 1$ structure: eight paired states plus a singleton $256 = 100000000$. The support is not dynamically closed: its image under the frozen transition lies outside the support. Therefore the object is best described as a localized event support, not a persistent orbit of the microscopic transition.

9 Results V: multi-observer and observation-method consensus

9.1 Frozen-transition multi-observer consensus

The frozen-transition multi-observer audit loads the saved transition table at iteration 8 and rebuilds the atlas 64 times with independent observer seeds. In all 64 reconstructions, the target namespace is observed and exact S_3 is recovered. Support-overlap matching also recovers exact S_3 with Jaccard overlap 1.0.

The summary reports:

- target exact S_3 consensus fraction: 1.0;
- support exact S_3 consensus fraction: 1.0;
- target support size: 17;

- maximum target/support generated group order: 6;
- maximum target/support transposition component size: 3;
- observer-dependent candidate: false.

This is the strongest evidence that the S_3 event is not merely one observer's chart artifact. Under the tested reconstruction family, the frozen transition forces the observer to recover the same nonabelian support.

9.2 Observation-method scale test

The observation-method scale test varies observer profile, atlas capacity, and chart entropy threshold over 48 tested configurations. Each configuration uses 16 observer reconstructions. All 48 configurations recover exact target and support-level S_3 , with support Jaccard 1.0, support generated group order 6, and maximum transposition component size 3.

The tested grid includes observer profiles `full_atlas`, `fiber_preserving`, and `boundary_only`; capacities 16, 32, 64, 128; and minimum chart entropy thresholds 0, 0.025, 0.05, 0.1, with fixed chart-coordinate settings in the reported small grid. The result should therefore be stated as invariance across this tested observation-method family, not across all possible observers.

10 Negative results and controls

Negative results are essential because they locate the actual mechanism.

10.1 Certified C_3/C_2 lateral interface is product-like

A certified-sector interface audit constructs a certified C_3 event sector and a certified C_2 frame sector. The certified interface exists in the twist/hybrid sectors, but it remains product-like. The summary reports certified-interface fraction 0.4, product-like fraction 0.4, compressed fraction 0, coupled-candidate fraction 0, product-subtracted candidate fraction 0, maximum period 6, and maximum bijective domain 6. Thus the side-by-side interface realizes the product baseline $C_3 \times C_2 \cong C_6$, not a semidirect S_3 extension.

10.2 Paired return and moving carriers do not produce exact S_3

Paired-return audits construct a shared carrier (σ, f) and test observed return maps. They find either empty carriers, size-two cores, weak feedback, or vacuous relations, but no order-three return, no noncommuting return, and no strict S_3 . Moving-carrier audits find transport arrows and weak C_2 naturality failure, but again no order-three moving return, no noncommuting cycle, and no strict S_3 .

10.3 Endpoint and schedule-cut crossing phases are negative

Endpoint and schedule-cut crossing-phase audits test whether a hidden pre/mid/post phase cycle yields S_3 . Both are negative. Phase rows exist and C_2 flips can be present, but no three-phase triplet, no aggregate order-three map, and no aggregate S_3 appear. This rules down the hypothesis that the nonabelian structure arises as a deterministic internal crossing phase.

10.4 $v = 10$ scaling control

The broad $v = 10$ run increases C_2 generation and final persistence, but its spatial closure remains a pure C_2 baseline: 4812 same-namespace C_2 groups, all single/duplicate flips, no shared-label pair, no component of size 3, and no exact S_3 . Thus the S_3 event is not a monotone consequence of more vertices or more C_2 abundance. It requires transposition diversity in a local namespace.

11 Interpretation: gauge, consensus, and event-like matter

The results support a distinction between three layers:

1. **Gauge baseline:** robust C_2 chart holonomy generated by information-preserving finite self-observation.
2. **Nonabelian reconciliation event:** a local critical window where two incompatible C_2 chart disagreements overlap, forcing exact S_3 .
3. **Matter-like consensus candidate:** a nontrivial support that independent observers recover as the same structure under support-level matching and observation-method variation.

The verified 17-state event satisfies the first two layers and passes the tested multi-observer consensus tests. It is therefore reasonable to call it a localized structural S_3 consensus event. It is not yet a particle in the usual physical sense because it has not been shown to propagate, recur, carry a conserved charge, or exhibit perturbation resistance. It is event-like rather than object-like.

This suggests the following operational definitions.

Definition 3 (Gauge disagreement). *Gauge disagreement is the groupoid generated by transition maps between distinct chart descriptions of the same boundary support.*

Definition 4 (Matter-like consensus). *A matter-like consensus support is a localized microstate support whose nontrivial reconciliation group is recovered by independent observer reconstructions over a specified observation-method class.*

Under these definitions, the reference 17-state support is the first matter-like consensus candidate in the framework.

Statistical status of the rare event

The spatial closure rate is deliberately reported as rare: one exact S_3 namespace among 3838 C_2 -bearing namespaces in the v9 chart-cycle closure audit, and one exact critical birth-window event among 1165 generated birth-window namespaces. The claim in this paper is therefore not a prevalence claim. Frequency alone is not used as evidence. The evidence is that the rare event has an exact finite-group certificate, a localized 17-state support, a matched-control contrast, 64/64 frozen-transition observer recovery, and 48/48 recovery in the reported observation-method grid. A future submission should add a constrained label-shuffle or loop-map-shuffle null distribution for the exact 1/3838 count; the present draft treats this as an explicit limitation rather than a completed significance test.

11.1 Aggregate certified-sector near-miss is not positive evidence

A separate certified-sector algebra audit reported a small aggregate “S3-like” C3/C2 interface signature without exact group proof. This is not used as evidence for the main result. The corresponding certified interface remains product-like: certified C3 and C2 sectors coexist on a six-state product baseline, with no compressed interface, no product-subtracted candidate, and no exact semidirect closure. This near-miss is useful as a negative control because it shows that the present positive S3 event is not a generic artifact of any six-state C3/C2 product table.

11.2 Paired-return verdict correction

Early paired-return verdict text overstated a subcritical feedback diagnostic. The corrected interpretation is that generated candidates can show size-two coordinate-feedback diagnostics, but no valid size-six joint return carrier, no order-three return, no noncommuting return, and no nonvacuous S3 relation. These paired-return rows are therefore negative controls, not supporting evidence.

12 Limitations

The present result is strong but limited.

1. **Existence, not prevalence.** The verified S_3 event is unique in the reported $v = 9$ birth-window audit. It is structurally verified but rare.
2. **Finite observation-method family.** The scale test covers 48 configurations, not all possible observers. The result should be stated relative to that tested family.
3. **No Standard Model identification.** S_3 is not $SU(2)$ or $SU(3)$. The result demonstrates finite nonabelian gauge reconciliation, not Standard Model recovery.
4. **No propagation yet.** The 17-state support maps out of itself and the local S_3 pulse lasts one tick. A memory-transfer or perturbation-stability audit is required before claiming particle-like dynamics.
5. **Implementation dependence.** Although the package provides reproducibility tests, any computational result depends on the correctness of the implementation. The paper mitigates this by exposing frozen transition tables, support hashes, synthetic tests, and simple finite-group certificates.

13 Reproducibility

Repository scope

The accompanying repository is a real computational package for the central witness and consensus tests. It includes the actual `relgauge` modules needed for the core positive-result chain, the frozen transition table `data/transition_iter8_inst28_seed3221741.npy`, the 17-state reference support metadata, shell scripts, tests, and the paper source. It does not rely on JSON-summary wrapper scripts for the central frozen-transition checks.

The repository is intentionally scoped. The script `reproduce.sh` runs the *core positive-result reproduction chain*: witness generation, regenerated-transition comparison, critical birth-window detection, frozen-transition multi-observer consensus, observation-method scale, and lift-mode

comparison. It does not rerun every exploratory or negative-control audit described in the manuscript. The broader negative controls—certified C_3/C_2 interfaces, paired returns, moving carriers, endpoint and schedule-cut crossing phases, broad $v = 10$ scaling controls, and other exploratory modules—are reported from archived run artifacts and are not all included in the minimal core package.

Package commands

The package uses Bash-style line continuation. A reviewer can run:

```
pip install -r requirements.txt
bash test_imports.sh
bash reproduce_quick.sh
```

The quick script recomputes the frozen-transition multi-observer and observation-method scale tests from the bundled transition table. It should reproduce the target/support-level S_3 consensus without reading precomputed summary JSON files.

For the longer core chain:

```
bash reproduce.sh
```

This can take tens of minutes to more than an hour depending on CPU and disk speed because it regenerates witness CSVs and chart-cycle rows. The package also includes `requirements-lock.txt` and `environment.yml` recording the validated Python and dependency versions.

14 Conclusion

This paper demonstrates a concrete finite route from binary self-observation to nonabelian gauge reconciliation. Information-preserving finite atlas iteration robustly generates C_2 chart holonomy. In a critical local birth/rebound window, two overlapping same-namespace C_2 flips appear on a 17-microstate support. Their closure is exact S_3 . Freezing the transition at that moment and reconstructing the atlas independently yields exact target and support-level S_3 in all tested observer reconstructions, and the result survives all tested observation-method configurations.

The result does not derive the Standard Model. It does not show that S_3 is generic. It does show that a finite binary boundary-observation pipeline can produce localized, structural, nonabelian gauge consensus without inserting a nonabelian group by hand. The next decisive tests are support-flux, transition-neighborhood consensus at iterations 7, 8, 9, and perturbation stability of the 17-state support.

A Core algorithms in pseudocode

This appendix gives implementation-level pseudocode for the operations used in the reference audits. The goal is not to replace the released code, but to make the mathematical construction explicit enough that a reader can reconstruct the finite objects independently.

A.1 Iterated finite self-observation

```
Input: finite state list S, transition table T:S->S, alphabet q,
      atlas capacity K, iteration count M, observer parameters theta.
Output: atlas summaries A_t, effective transitions T_t.

T_0 := T
for t = 0,...,M:
```

```

A_t := BUILD_ATLAS(S, T_t, theta)
L_t := PROFILE_LABELS(A_t, T_t, profile)
B_t := COMPRESS_LABELS_TO_CAPACITY(L_t, K)
R_t := TEMPORAL_RELATION(B_t, T_t)
T_{t+1} := INFORMATION_PRESERVING_LIFT(B_t, R_t, S, T_t)
record domains, atlas maps, chart cycles, C2/S3 holonomies
end for

```

The reference experiment uses a deterministic seed schedule. Python’s process-randomized `hash()` is not used; profile-dependent seeds are computed by a stable SHA1 hash. When `--save-transition-at t` is enabled, the array representing T_t is written to disk before the atlas pass at iteration t .

A.2 Atlas generation and recursive proliferation

```

BUILD_ATLAS(S, T, theta):
  D_0 := { root domain with initial boundary label pi_0(s) }
  D := D_0
  for depth = 1,...,max_depth:
    new_domains := []
    for domain d in D at previous depth:
      for each live fiber F of d with |F| >= min_fiber_size:
        compute predictive signatures of states in F under T over horizon h
        split F by signatures satisfying entropy/live-class filters
        keep at most max_domains_per_depth highest-scoring child domains
        append kept children to new_domains
    D := D union new_domains
  return domains D plus chart data from BUILD_CHARTS(D,T,theta)

```

A domain is therefore not a spatial region in a pre-existing space. It is a finite predictive partition induced by the boundary projection and the transition table. A fiber is a set of microstates sharing the same domain label.

A.3 Chart construction on one fiber

```

BUILD_CHARTS_FOR_FIBER(domain d, fiber F, state table S, transition T):
  charts := []
  for coordinate subset C with |C| <= max_chart_coords:
    label_C(s) := tuple of selected boundary/support coordinates of s
    if label_C partitions F into at least min_chart_classes
      and entropy(label_C on F) >= min_chart_entropy
      and every retained class has enough support states:
        add chart c=(C,label_C,support=F)
  keep at most max_charts_per_fiber charts by entropy/support score
  return charts

```

The concrete implementation uses deterministic enumeration or seeded deterministic selection of coordinate subsets subject to the same filters. Changing these admissible chart parameters is what the observation-method scale audit tests.

A.4 Overlap maps and chart-cycle holonomy

```

OVERLAP_MAP(chart c_i, chart c_j):
  shared := support(c_i) intersection support(c_j)
  if |shared| < min_overlap_states: reject
  for each source label a of c_i on shared:
    count target labels b = c_j(s) over states s with c_i(s)=a
    if one target b has deterministic majority passing threshold:
      map a -> b
  else:
    mark source label nondeterministic

```

```

return partial map g_ij on deterministic source labels

CHART_CYCLE_HOLONOMY(charts, overlap maps):
  build graph whose vertices are charts and edges are valid overlap maps
  enumerate simple cycles up to max_cycle_len
  for cycle c0->c1->...->ck=c0:
    H := g_{c_{k-1},c_k} o ... o g_{c_0,c_1}
    core := largest subset of labels closed under H and H^{-1}
    if H restricted to core is a permutation:
      order := permutation_order(H|core)
      record C2 if order=2; record loop map string

```

The same-namespace rule is applied after this step: two loop maps are composed for group closure only when they belong to the same candidate, atlas iteration/window, parent domain, and fiber label, or when a later support-overlap consensus test certifies that two observers are describing the same microstate support.

A.5 Transposition diversity and exact S_3 closure

```

TRANSPPOSITION_DIVERSITY(namespace N):
  maps := distinct order-2 loop maps recorded in N
  E := set of single-transposition edges {a,b} induced by maps
  build graph G_N with label vertices and transposition edges E
  component_size := size of largest connected component of G_N
  generated_group := closure of maps as finite permutations on union core
  exact_S3 := (|generated_group|=6 and group is nonabelian
               and component_size=3)
  return distinct map count, component_size, exact_S3

```

For the reference event, $E = \{\{0, 1\}, \{0, 2\}\}$, so the generated group is exactly S_3 .

A.6 Critical birth-window provenance

```

CRITICAL_BIRTH_WINDOW(namespace N at iteration t):
  first_window := |t - first_C2_iteration(candidate)| <= 1
  c2_birch := (C2_count(t) > 0 and C2_count(t-1) = 0)
  signature_delta := full_signature_classes(t) - full_signature_classes(t-1)
  fiber_entropy_delta := fiber_entropy(t) - fiber_entropy(t-1)
  valid_cycle_drop := max(0, valid_cycles(t-1)-valid_cycles(t)) / max(1, valid_cycles(t-1))
  rebound_score := max(0, signature_delta
                       + 50*max(0, fiber_entropy_delta)
                       + 100*valid_cycle_drop
                       + (25 if c2_birch else 0))
  priority := 5*transposition_diversity_score + 0.01*rebound_score
             + domain-depth/live-fiber/predictive-class terms
  return provenance fields and exact_S3 classification

```

The rebound score is a heuristic provenance statistic, not a fitted physical observable. It is used to compare the verified S_3 event to matched C_2 controls and to prioritize near misses.

A.7 Multi-observer and support-overlap consensus

```

MULTI_OBSERVER_CONSENSUS(frozen transition T*, reference support R):
  for observer seed/configuration omega in tested family:
    A_omega := BUILD_ATLAS(S, T*, theta_omega)
    groups := all same-namespace transposition-diversity groups in A_omega
    exact_target := group at target parent/fiber is exact S3, if present
    for each group g in groups:
      J(g,R) := |support(g) intersection R| / |support(g) union R|
      overlap(g,R) := |support(g) intersection R| / |R|

```



```
best := group maximizing J(g,R), subject to overlap thresholds
record whether best is exact S3, C2-only, or flat
report fractions over observer seeds/configurations
```

The term “multi-observer” in this paper therefore means algorithmic stability across a specified reconstruction family. It is a finite model of observer consensus, not a proof of observer-independence in the physical or philosophical sense.

B Reproduction commands

B.1 Reference witness generation

```
python -m relgauge.iteratedfiberatlasdynamicsaudit 2 \
--vertices 9 \
--rule-modes random_full_permutation \
--instances 29 \
--atlas-capacities 32 \
--profiles full_atlas \
--atlas-iterations 12 \
--proliferation-iterations 4 \
--horizon 3 \
--max-state-samples 512 \
--save-transition-at 8 \
--seed 2000006 \
--out results/repro_witness_search.csv
```

B.2 Critical birth-window audit

```
python -m relgauge.criticalc2birthwindowaudit \
--chart-cycles-csv results/repro_witness_search_chart_cycles.csv \
--iterated-csv results/repro_witness_search.csv \
--domains-csv results/repro_witness_search_domains.csv \
--require-generated --require-flat-start \
--focus-first-c2-window 1 \
--namespace-cols parent_domain_id,fiber_label \
--out results/critical_birth.csv
```

B.3 Multi-observer consensus

```
python -m relgauge.multiobserverconsensusaudit 2 \
--vertices 9 \
--iterated-csv results/repro_witness_search.csv \
--frozen-transition-npy data/transition_iter8_inst28_seed3221741.npy \
--target-rule-mode random_full_permutation \
--target-instance 28 \
--target-profile full_atlas \
--target-atlas-capacity 32 \
--target-seed 3221741 \
--target-iteration 8 \
--target-parent-domain 75 \
--target-fiber-label 7 \
--observer-runs 64 \
--max-state-samples 512 \
--out results/multi_observer_consensus.csv
```

B.4 Observation-method scale test

```
python -m relgauge.observationmethodscaletest 2 \
--vertices 9 \
--iterated-csv results/repro_witness_search.csv \
--frozen-transition-npy data/transition_iter8_inst28_seed3221741.npy \
--target-rule-mode random_full_permutation \
--target-instance 28 \
--target-profile full_atlas \
--target-atlas-capacity 32 \
--target-seed 3221741 \
--target-iteration 8 \
--target-parent-domain 75 \
--target-fiber-label 7 \
--observer-profiles full_atlas,fiber_preserving,boundary_only \
--observer-atlas-capacities 16,32,64,128 \
--min-chart-entropy-list 0,0.025,0.05,0.1 \
--observer-runs 16 \
--max-state-samples 512 \
--out results/observation_method_scale.csv
```

C Reference support hash

The reference support is

$$\{29, 30, 44, 47, 61, 62, 156, 159, 188, 191, 221, 222, 253, 254, 256, 284, 287\}.$$

The support hash reported by the multi-observer audit is

$$\text{ffb7d91caf4970bcfd60}.$$

D Recommended future audits

1. **Transition-neighborhood consensus.** Save and test transitions at iterations 7, 8, 9 to verify that structural S_3 consensus is localized to iteration 8.
2. **Support-flux memory audit.** Track the image of the 17-state support and test whether it overlaps later gauge-bearing supports.
3. **Perturbation stability.** Modify transition rows inside/outside the support and measure the minimum perturbation that destroys S_3 consensus.
4. **Broader observation-method grid.** Vary chart-coordinate limits, support-coordinate limits, and charts per fiber beyond the initial 48 configurations.

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Table 1: Principal computational artifacts.

Artifact family	Role
Witness lift-mode summaries	Reference 29-instance $q = 2$, $v = 9$, full-atlas, capacity-32 lift-mode comparison.
Broad $v = 9$ iterated atlas run	Gauge-generation and spatial-closure source run.
Broad $v = 10$ iterated atlas run	Scaling/control run showing strong C_2 generation but no S_3 closure.
Spatial closure and transposition-diversity audits	Same-namespace composition of C_2 chart loops; detects one exact S_3 event in $v = 9$.
Critical birth-window audit	Birth-window provenance; detects one S_3 critical window among C_2 birth-window namespaces.
Critical-event microscope	Group/orbit certificate and matched C_2 controls.
Critical pulse train	Temporal localization of the S_3 event as a one-tick flash.
Frozen multi-observer consensus	64-run exact/support S_3 consensus on the frozen transition.
Observation-method scale test	48-configuration observation-method scale test.

Table 2: Reference lift-mode result. The representative lift is information-destroying and does not generate gauge in this test.

Lift mode	C_2 generation	Final C_2	Max C_2 holonomy	Interpretation
bijjective	0.689655	0.0	116	information-preserving
permutation	0.689655	0.0	116	information-preserving
conservative	0.689655	0.0	116	information-preserving in this regime
information_conserving	0.689655	0.0	116	information-preserving
representative	0.0	0.0	0	information-destroying

Table 3: Critical-event microscope: target versus matched C_2 controls.

Class	Count	Group order	Largest orbit	Mean rebound score
Target S_3 event	1	6	3	359.50
Matched C_2 controls	20	2	2	25.05